



Heat and mass transfer during evaporation of thin liquid films confined by nanoporous membranes subjected to air jet impingement

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ABSTRACT

Thin liquid films can provide efficient means for heat and mass transfer by supporting a large rate of evaporation, while maintaining temperatures under saturation conditions. The use of nanoporous membranes establishes thin liquid films minimizing the possibility of dryout by exploiting capillary confinement of the fluid. In combination with gas jet impingement, this yields record high heat and mass fluxes. Consequently, for applications relying on efficient heat and mass transfer, such as, phase change cooling of electronics and membrane distillation of water, it is essential to identify and characterize all the underlying transport mechanisms affecting overall performance. A detailed computational analysis of heat and mass transfer is therefore carried out to predict the performance of thin film evaporation. Heat conduction and convection are both found crucial in determining the temperature distribution within the thin film region, which ultimately controls the rate of evaporation from the liquid–vapor interface. Furthermore, the effect of vapor transport from the liquid–vapor interface on overall performance is also assessed. The relative effects of these performance-governing parameters are discussed and quantitatively compared at different operating conditions. The results of the computational analysis are supported by comparison with experiments.

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1. Introduction

Thin liquid films can provide efficient heat and mass transfer by taking advantage of both the conductive/convective and phase-change modes of heat transfer, while maintaining the maximum operating temperature under the saturation conditions. Phase change heat transfer can be significantly improved if the thermal resistance across the evaporating thin liquid film is minimized, while the vapor transport from the liquid–vapor interface is maximized. Consequently, evaporation of thin liquid films has generated significant interest [1–12]. Cooling strategies that promote formation of thin liquid films include mist generation [13], gravity [14–16] and gas assisted thinning [17–19] of free surface flows. Typically, gas-assisted thin film evaporation utilizes a high velocity flow of liquid–gas mixture within a narrow gap between surfaces dissipating heat [17,18]. Similarly, shear-thinning of liquid films caused by two-phase flow [19] has also been demonstrated for surface cooling via evaporation. The critical drawbacks that limit the

overall cooling performance of these mechanisms are the occurrence of dryout and the lack of precise control over the film thickness due to reliance on hydrodynamics. Alternatively, capillary action can also be utilized to drive liquid and maintain a thin film, which is integral to heat pipes [7,20–23]. The maximum heat dissipation in this case is controlled by the capillary pumping limit. Capillary confinement of liquids using nanoporous membranes has been successfully demonstrated to yield precise control over film thickness, resulting in dissipation of very high heat fluxes for electronics cooling applications [24].

Liquid-to-vapor phase change by evaporation is not limited to thermal management applications. If carried out efficiently, evaporation can also be used to extract distillates by ensuring large mass transfer rates. This is desirable for applications such as distillation or purification of water. In particular, membrane distillation is an emerging technology for water purification and desalination, wherein the transport of water–vapor across a porous membrane is thermally driven by a gradient in the vapor pressure [25–28]. For applications relying on optimal heat and mass transfer from evaporating films, it is essential to identify and characterize the physics of underlying transport mechanisms and to define the rate limiting process.

In this paper, a detailed computational analysis of heat and mass transfer in an evaporating water film in a micro-scale device

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Nomenclature

A	Area (m^2)
Br	Brinkman number
$\bar{c}$	Intercept in linear regression model for RTD temperature calibration (K)
c_p	Specific heat (J/kg K)
d_h	Hydraulic diameter (m)
D_k	Coefficient of Knudsen diffusion (m^2/s)
h	Heat transfer coefficient (W/mK)
h'	Mass transfer coefficient (m/s)
$\hat{h}_{fg}$	Latent heat of vaporization (J/kg)
k	Thermal conductivity (W/mK)
$\bar{m}$	Slope in linear regression model for RTD temperature calibration (K/Ω)
M	Molecular weight (kg/kmol)
$\dot{m}''$	Mass flux ($\text{kg/m}^2 \text{ s}$)
$\dot{m}'''$	Volumetric rate of evaporation ($\text{kg/m}^3 \text{ s}$)
p	Pressure (N/m^2)
Pr	Prandtl number
$\dot{q}''$	Heat flux (W/m^2)
$\dot{q}'''$	Volumetric heat generation (W/m^3)
R	Electrical resistance (Ω)
$\bar{R}$	Universal gas constant (J/kmol K)
T	Temperature (K)
$\bar{u}$	Velocity (m/s)
V	Voltage (V)
$\bar{V}$	Volume (m^3)
X	Mass fraction

Greek symbols

δ	Thickness (m)
ε	Membrane porosity
μ	Dynamic viscosity (kg/ms)
ρ	Density (kg/m^3)
$\hat{\sigma}$	Accommodation coefficient for evaporation or condensation at the interface

Subscripts

a	Air
al	Alumina
atm	Atmospheric
f	The interface of liquid and membrane in the computational analysis
hs	Heater surface
∞	Ambient conditions surrounding the membrane
l	Liquid
l,in	Liquid at the device inlet
li	Liquid surrounding the interface
m	Membrane
p	Pyrex
pr	Precision resistance
s	Silicon
v	Vapor
v,eq	Vapor in equilibrium with the interface
vi	Vapor surrounding the interface
vo	Vapor at the membrane outlet

enabling capillary confinement of thin liquid film is presented and simulation results are discussed with support of relevant experimental data. It is shown how heat conduction (for the stagnant film) or convection (due to flow of the liquid) govern the temperature distribution within the thin film region, and how these factors ultimately control the rate of evaporation from the liquid–vapor interface for liquid films with $>1 \mu\text{m}$ thickness. Furthermore, it is also shown that vapor transport from the liquid–vapor interface is essential to enhance overall performance in applications relying on evaporation. The computational analysis is supported by results from experiments carried out at similar operating conditions. The results presented herein, although derived from the study of a specific micro-scale device enabling thin film evaporation, are generally applicable to systems that rely on efficient heat and mass transfer resulting from evaporation of confined thin liquid films.

2. Experimental system enabling investigation of controlled film thickness evaporation

The evaporating liquid is supplied through a narrow channel bounded by a heat generating surface, labeled as heater, and a nanoporous alumina membrane [24], as shown in Fig. 1. Being inherently hydrophilic, the porous alumina membrane drives the liquid into the nanopores by capillary action against the viscous drag resisting flow through the pores. During evaporation, when the viscous shear exceeds the pressure differential driving liquid across the membrane, a complete saturation of the membrane with liquid is prevented and the liquid–vapor interface is restrained between the inlet and outlet of the membrane pores. Therefore, the membrane pores are generally only partly filled with the liquid, the rest being occupied by the air–vapor mixture. A detailed study of interfacial transport during evaporation of water confined within nanostructured surfaces was previously carried out [29] to elu-

cidate the effect of electrostatic interaction, in addition to capillary and disjoining pressures. In this case, the rate of heat conduction across the liquid–film is limited only by the film thickness, which is governed by the separation of the membrane from the heater – a parameter, which can be precisely controlled using micro-fabrication [24]. On the other hand, if the pores of the alumina membrane are rendered hydrophobic with a suitable chemical coating, then the liquid–vapor interface will be pinned at the membrane pore inlet itself and the entire pore will then be occupied only by the air–vapor mixture. In such a case, a gas-phase mass transfer of vapor within the pores may become the rate limiting process if the pore aspect ratio (length-to-diameter) is sufficiently large [13].

For a hydrophilic membrane, due to the variable pinning location of the liquid–vapor interface within the pore, the extent of liquid penetration within the membrane can be quite different across the membrane with near cylindrical nanopore structure (Fig. 1b). Consequently, for a hypothetical case, wherein a membrane pore is fully saturated with liquid, the net vapor transport will be limited only by advection due to the air stream at the membrane outlet. On the other hand, for a membrane void of liquid, as in the case of a hydrophobic membrane, the vapor resulting from evaporation at the liquid–vapor interface must first diffuse across the entire length of the pore before being transported via advection at the membrane outlet. The net vapor transport is then limited by diffusion as well as advection. Therefore, these two limiting cases set the overall range in the variation of net vapor transport, provided other operating conditions remain invariant.

With liquid trapped within the hydrophilic membrane, evaporation occurs due to the heat generated at the heater. This may result in significant cooling and high vapor flux. Furthermore, heat dissipation resulting from the liquid flow across the heater can also contribute towards overall cooling. It is apparent that for liquid flow rates much larger than the overall rate of evaporation, the

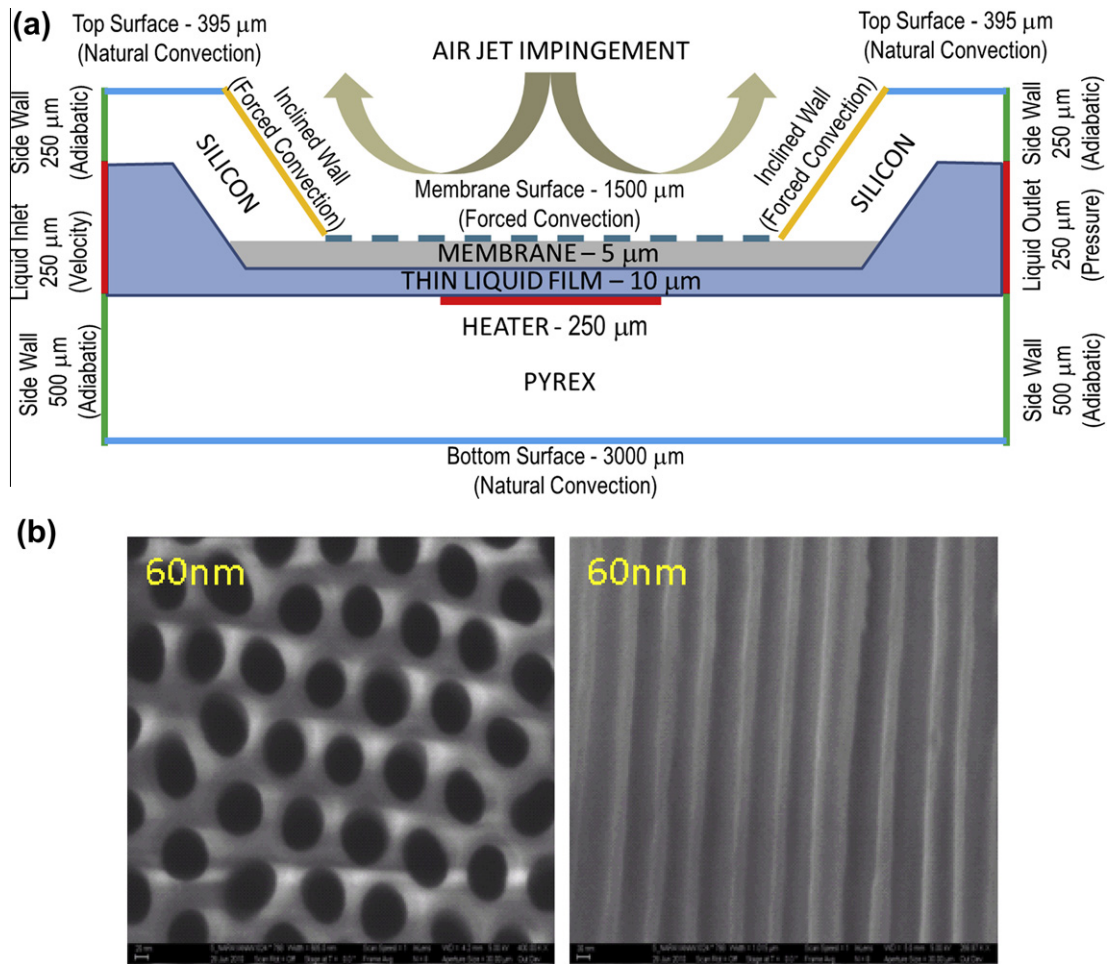


Fig. 1. (a) A microscale device to enable thin film evaporation. The physical model illustrates key geometric parameters and boundary conditions for the analysis of heat and mass transfer. (b) The magnified SEM views of the membrane that confines the liquid over the heated surface. The top and side views illustrate the cylindrical geometry of the porous alumina (PA) pores.

capacitive heating of liquid between the porous membrane and the heated surface can be significant. When both capacitive and evaporative cooling are not effective, heat rejection and spreading from the heater into the substrate will be prominent. Therefore, for any given liquid flow rate and heat generation at the surface, this analysis compares the rate of heat dissipation due to evaporation, single-phase cooling and the extent of spreading into the substrate. The temperature distribution across the heated surface and the membrane is predicted in order to estimate the thermal resistances across the liquid film, membrane, and the region surrounding the membrane-outlet. The computational predictions are supported by experiments carried out at similar operating conditions.

3. Mathematical model

3.1. Equations governing fluid flow and heat transfer

The characteristic lengths of the heater (250 μm) and the membrane (1500 μm) are orders of magnitude larger than the liquid-film and membrane thicknesses (10 μm and 5 μm, respectively). Due to the characteristic geometry of the flow cross section (liquid film thickness $\ll$ out-of-plane channel depth), out-of-plane fluid flow can be neglected in comparison to the in-plane fluid flow. Therefore, a two-dimensional analysis of flow and heat transfer is carried out to predict the rate of heat dissipation from the heater for different operating conditions. Assuming a Newtonian fluid

with constant density, ρ_l and viscosity, μ_l , the steady flow of an incompressible liquid confined within the microchannel is governed by the following equations:

$$\nabla \cdot \bar{u}_l = 0 \quad (1)$$

$$\rho_l \bar{u}_l \cdot \nabla \bar{u}_l = -\nabla p_l + \mu_l \nabla^2 \bar{u}_l \quad (2)$$

where $\bar{u}_l$ and p_l represent the velocity and pressure of the liquid. The temperature distribution, T_l within the liquid is given by the following equation, which assumes constant thermal conductivity, k_l , no internal heat generation and negligible fluid compressibility effects

$$\rho_l c_{pl} \bar{u}_l \cdot \nabla T_l = k_l \nabla^2 T_l \quad (3)$$

It is to be noted that viscous dissipation due to flow in the narrow channel is also neglected, since the Brinkman number, $Br = u_l^2 Pr_l / c_{pl} \Delta T_l$, for the typical operating conditions is found to be negligibly small $\sim 10^{-9}$, where u_l represents the characteristic velocity scale. The temperature distributions in the pyrex (T_p) and silicon (T_s) substrates (Fig. 1) are governed by the following equations, respectively:

$$k_p \nabla^2 T_p + \dot{q}_j''' = 0 \quad (4)$$

$$\nabla^2 T_s = 0 \quad (5)$$

where, k_p is the thermal conductivity of pyrex (assumed constant), and $\dot{q}_j'''$ represents volumetric heat generation within the substrate, generated by the heating element.

3.2. Evaporation within the membrane

In order to determine the rate of evaporation inside the membrane pores, the vapor generated at the interface and its transport via diffusion and advection are quantified. The mass flux due to evaporation at the liquid–vapor interface surrounded by an air–vapor mixture is given by [30]:

$$\dot{m}'' = \frac{2\hat{\sigma}}{2 - \hat{\sigma}} \left(\frac{M_l}{2\pi R} \right)^{1/2} p_{atm} \left(\frac{X_{v,eq}}{T_{li}^{1/2}} - \frac{X_{vi}}{T_{vi}^{1/2}} \right) \quad (6)$$

$X_{v,eq}$ and X_{vi} are the equilibrium and interfacial mole fractions of vapor phase, defined as $p_{v,eq}/p_{atm}$ and p_{vi}/p_{atm} , respectively, where $p_{v,eq}$ and p_{vi} denote the saturated vapor pressure at temperature T_{li} and the partial pressure of vapor in the air–vapor mixture in the immediate vicinity of the interface, respectively. T_{li} and T_{vi} represent the liquid and vapor temperatures in the immediate vicinity of the interface, respectively. M_l is the molecular weight of the evaporating liquid.

The rates of vapor transport by mass diffusion inside the membrane pores and by convection due to the sweep air at the membrane outlet, are given by Eqs. (7) and (8), respectively

$$\dot{m}'' = \frac{D_k M_l p_{atm}}{\delta_v R} \left(\frac{X_{vi}}{T_{vi}} - \frac{X_{vo}}{T_{vo}} \right) \quad (7)$$

$$\dot{m}'' = \frac{h' M_l p_{atm}}{R} \left(\frac{X_{vo}}{T_{vo}} - \frac{X_{v\infty}}{T_{\infty}} \right) \quad (8)$$

where X_{vo} and T_{vo} represent the mole fraction and temperature of the vapor phase at the membrane outlet, respectively; D_k represents the coefficient of combined Knudsen–Molecular diffusion [13] across the length δ_v , which denotes the length of vapor column inside the pore; h' is the mass transfer coefficient for the transport of the vapor phase by air with far-field coolant vapor mole fraction $X_{v\infty}$ and temperature T_{∞} . Based on the location of the interface inside the membrane pore, the possible values of δ_v are given by $0 \leq \delta_v \leq \delta_m$, where δ_m is the thickness of the membrane.

Assuming the vapor and membrane to be in local thermal equilibrium, the analysis can be simplified by neglecting the temperature difference, $T_{vi} - T_{vo}$. This assumption is justified for thin ceramic (alumina) membranes ($\delta_m \sim 10^{-6}$ m) with thermal conductivity, $k_m \sim 10$ W/mK, since the temperature difference across the membrane is less than 0.1 K even for heat fluxes as high as 10^6 W/m². The liquid and vapor at the evaporating interface are assumed to be in thermal equilibrium, so that $T_{li} = T_{vi}$ [29]. In other words, the difference in temperatures, $T_{vi} - T_{vo}$ and $T_{li} - T_{vi}$ are neglected whereas the differences in mole-fractions $X_{vi} - X_{vo}$ and $X_{v,eq} - X_{vi}$ are still considered significant. This simplification is justified because the temperature discontinuities are rendered negligible by the efficient thermal transport across the solid phase of the ceramic membrane. On the other hand, the differences in mole fractions are not negligible since vapor transport via diffusion is much less efficient.

Using a completely dry sweeping air ($X_{v\infty} = 0$), the overall rate of evaporation per unit (footprint) area, after including the above stated simplifications, is given by the following equation, which takes into account the available area for evaporation via the membrane porosity, ε :

$$\dot{m}'' = \frac{p_{atm} M_l X_{v,eq} \varepsilon}{RT_{vi}} \left(\frac{1}{(2 - \hat{\sigma}) / (2\hat{\sigma}) (2\pi M_l / (RT_{vi}))^{1/2} + \delta_v / D_k + 1/h'} \right) \quad (9)$$

Heat dissipation by evaporation within the thin porous membrane is approximated by a uniform volumetric heat sink term, $\dot{q}_m'''$, which is included in energy conservation equation for the membrane (10). The volumetric heat sink is derived from heat and mass transfer balance for the rate of evaporation inside the membrane as,

Table 1

Thermophysical properties of the working fluid and substrates.

	Density (kg/m ³)	Conductivity (W/mK)	Viscosity (kg/ms)
Water	998.2	0.6	1.003×10^{-3}
Liquid saturated membrane	2010.2	12.6	–
Vapor saturated membrane	1415	12.7	–
Pyrex	2230	1.1	–
Silicon	2324	66	–

$\dot{q}_m''' = \dot{m}'' \hat{h}_{fg} / \delta_m$. This allows coupling between the modes of heat transfer occurring within and outside the membrane. It is to be noted that the use of a volumetric heat sink term, as opposed to a uniform heat flux as a boundary condition in our analysis is pertinent to account for the variability in the location of the liquid–vapor interface inside the membrane. Furthermore, due to extremely small evaporation rates inside the membrane, convective heat transfer due to liquid and vapor flow in membrane pores is neglected in comparison to the latent heat of phase change and conduction across the solid structure of the membrane. The temperature distribution in the membrane can hence be given by:

$$k_m \nabla^2 T_m + \dot{q}_m''' = 0 \quad (10)$$

where k_m represents the volumetrically averaged thermal conductivity of membrane, which is given by $(1 - \varepsilon)k_{al} + \varepsilon k_l$ for a liquid-saturated membrane and by $(1 - \varepsilon)k_{al} + \varepsilon k_v$ for a dry, vapor-saturated membrane; k_{al} , k_l and k_v are the thermal conductivities of alumina, liquid and vapor phases, respectively. Computational analysis of the problem for the domain shown in Fig. 1 is carried out by solving the governing Eqs. 1, 2, 3, 4, 5, and (10) using a commercial computational fluid dynamics software, FLUENT [31]. The thermophysical properties of the liquid, Silicon and Pyrex substrates used in this analysis are given in Table 1.

3.3. Boundary conditions

The velocity at the “Liquid Inlet” (Fig. 1), $u_{l,in}$ is assumed to be uniform and assigned a value corresponding to the flow rates realized in the experimental study. The temperature of the liquid at the inlet is taken to be at the ambient condition, $T_{\infty} = 298$ K. Since the liquid exiting the device is exposed to ambient conditions, a constant gage pressure, $p = 0$ is assigned at the “Liquid Outlet” (Fig. 1). The heated surface denoted as “Heater” in Fig. 1 is computationally analyzed as a thin layer with uniform volumetric heat generation. The surfaces denoted as “Top Surface” and “Bottom Surface” in Fig. 1, are each assigned convective boundary conditions, with a heat transfer coefficient, $h_{\infty} = 10$ W/m² K, typical to natural convection under these conditions. The boundaries called “Side Walls” in Fig. 1, dissecting the region of interest from the overall device, are assigned the adiabatic boundary condition, representing walls that contribute negligibly to overall heat dissipation. The surfaces indicated as “Membrane Surface” and “Inclined Wall” in Fig. 1 are assigned convective boundary conditions with heat transfer coefficients h_m and h_s , respectively. The spatially varying heat transfer coefficients, h_m and h_s on these surfaces are determined from a separate computational analysis, detailed later, which predicts the flow resulting from jet impingement of air within the confined space at the membrane outlet.

The mass flow rate of the liquid confined between the membrane and the heated surface gradually decreases downstream as a result of evaporation. The change in flow rate is accounted in the analysis by including the loss in liquid, assuming a volumetric mass-sink in the computational cells at the boundary separating

the membrane and the liquid. The volumetric sink for the liquid adjacent to the membrane is given by the following equation

$$\dot{m}''' = -\dot{m}'' A_{f,cell} / \bar{V}_{cell} \quad (11)$$

$A_{f,cell}$, $\bar{V}_{cell}$ represent the face area and the volume of the computational cell, respectively and the evaporation mass flux $\dot{m}''$ is given by Eq. (9).

3.4. Computational analysis of air jet impingement

In order to determine the heat and mass transfer coefficients along the “Membrane Surface” and the “Inclined Wall”, a confined air jet, incident normally on the porous membrane is computationally analyzed, for the physical model illustrated in Fig. 1. In this two-dimensional analysis, the effect of air-jet confinement and thermal spreading in the substrates are studied on the overall rate of heat and mass transfer. The analysis of airflow is also carried out using FLUENT to determine the heat and mass transfer coefficients. Assuming a constant density and viscosity for air and a steady, laminar flow, numerical solutions of mass continuity, force momentum balance, and energy governing the flow of the air jet are carried out. The analysis assumes a constant thermal conductivity for air, with no internal heat generation, and neglects viscous heat generation and compressibility effects. The key results obtained from the analysis of air-flow are discussed in the [Supplementary information](#).

The local heat transfer coefficient, h_m , on the surface of the membrane is evaluated from the local surface heat flux dissipated by air, $\dot{q}_a''$ as $h_m = \dot{q}_a'' / (T_{vo} - T_\infty)$, where T_{vo} and T_∞ are the temperatures at the surface of the membrane and ambient, respectively. Note that the membrane surface and vapor outlet temperatures are the same due to local thermal equilibrium. The heat flux is given by $\dot{q}_a'' = -k_a \nabla T \cdot \hat{n}$, where the temperature gradient is obtained along the surface outward normal, $\hat{n}$. The local mass transfer coefficient, h' , along the membrane surface is evaluated from the heat transfer coefficient, h_m using the analogy between heat and mass transfer [32].

Numerical convergence is ensured by monitoring the net rate of heat transfer from all surfaces to check the global energy conservation and ensuring all residuals are less than 10^{-9} . Fig. 2 shows the effect of grid size on the computed values of heat transfer coefficient using air jet impingement. It is evident that the profile of heat transfer coefficient over the membrane surface is invariant to mesh

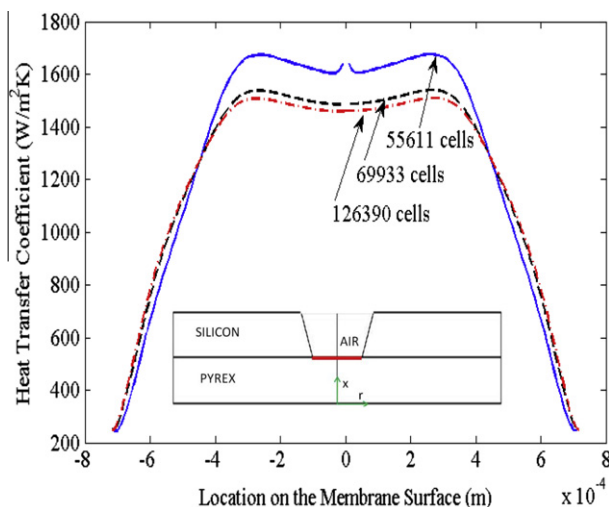


Fig. 2. Grid dependence study for air jet impingement illustrating the effect of grid size on the calculated magnitude of local heat transfer coefficient.

refinement from 69,933 to 126,390 cells, which translates to a minimum discrete element lengths of 4 and 2 μm , respectively.

4. Experimental study of evaporation

4.1. Micro-scale device for thin-film evaporation

A micro-scale device as shown in Fig. 3, fabricated using surface micromachining [24], constitutes an array of resistance temperature detectors (RTDs) and heaters, which simulate heated regions and also provide an accurate thermal map of the entire device surface during operation. The device also includes microchannels and ports for delivery of liquid to the heater. More importantly, the device houses a nanoporous ceramic membrane, which is monolithically integrated to eliminate unwanted contact resistances. The membrane confines a thin film over the heated surface and provides a passage for the vapor phase generated by evaporation to diffuse into the ambient as described earlier. The unique fabrication process allows variation in the geometric features of the microscale device. Specifically, the lateral dimensions of the heater, the thickness of the liquid film confined between the membrane and the heated surface can be altered in addition to variation in geometric features of the membrane such as porosity, thickness and lateral dimensions.

4.2. Experimental procedure

The performance of dry air-assisted evaporation is measured at different combinations of liquid and air flow rates to determine the relative effect of single phase convection and vapor advection, respectively. These effects are studied using devices with different characteristic dimensions of the film and membrane, while the heater dimensions in all test cases presented herein are $250 \times 25 \mu\text{m}^2$. The calibration of each RTD on the device is carried out prior to the experimental testing, and it provides a linear correlation of each RTD's resistance response to temperature [24]. The heater is activated using a DC power supply, causing Joule heating at the location of the heater where the resistance is highest. The supply of air and liquid are then initiated to enable evaporation of thin liquid films and heat rejection to the ambient.

Jet impingement of air is implemented using nozzles of different internal diameters (500 μm and 1000 μm) connected to a compressed air supply. The air supply line is connected in series with a flow meter and a pressure transducer to measure the volumetric flow rate and gage pressure, respectively. The nozzle for air jet impingement is held using a positioning tool such that its separation from the heated surface and the angle of impingement can be controlled. The results presented herein correspond to normal incidence of an air jet at different flow rates. The liquid used for thin film evaporation was de-ionized and de-aerated water, which was circulated using a syringe pump capable of precisely delivering flow rates as low as $\sim 10^{-10}$ to $10^{-8} \text{ m}^3/\text{s}$ ($\pm 0.5\%$). The temperature of the liquid at the inlet was measured using a calibrated thermocouple. The experiments were carried out at different air and liquid flow rates to determine the overall heat dissipation achievable via evaporation and sensible cooling.

4.3. Uncertainty analysis

The temperature at the heater is calculated based on the resistance values derived from the voltage and current measurements as shown in Fig. 4. The resistance at the heater, R_{hs} is given by $V_{hs} R_{pr} / V_{pr}$, where V_{hs} and V_{pr} represent the voltage drops measured across the heater and the precision resistor, respectively, and R_{pr} is the precision resistor ($3 \pm 0.03 \Omega$). The uncertainty in calculation of

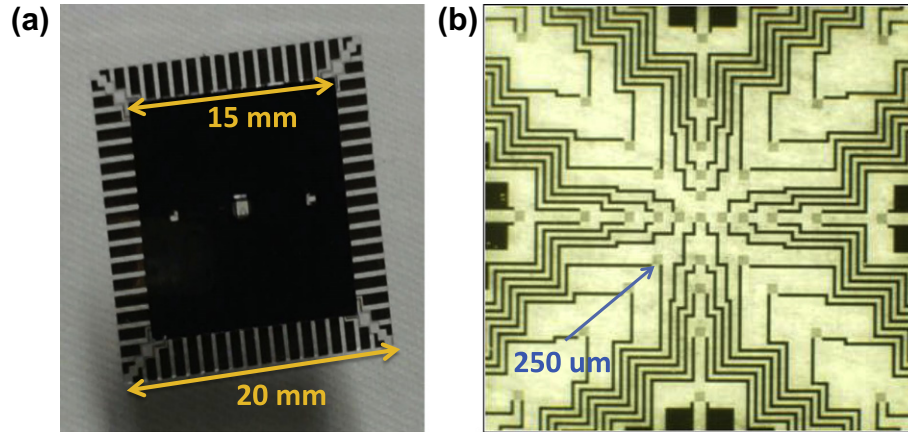


Fig. 3. (a) A photograph of microscale device enabling study of thin film evaporation in confined environment, with a centrally located heater and a nanoporous alumina membrane [24]. (b) The design and layout of RTDs and heater on Pyrex substrate.

the heater resistance due to the voltage measurements and to the precision resistance is given by the following equation:

$$(\Delta R_{hs}/R_{hs}) = \left((\Delta V_{hs}/V_{hs})^2 + (\Delta V_{pr}/V_{pr})^2 + (\Delta R_{pr}/R_{pr})^2 \right)^{1/2} \quad (12)$$

The maximum uncertainty in determining the heater resistance was therefore found to be less than 1%.

The temperature and heat flux at the heater are calculated as shown below:

$$T_{hs} = \tilde{m}R_{hs} + \tilde{c} \quad (13)$$

$$\dot{q}_{hs}'' = \frac{V_{hs}V_{pr}}{R_{pr}A_{hs}} \quad (14)$$

where A_{hs} represents the area of the heater. The slope, $\tilde{m}$, and intercept, $\tilde{c}$, are obtained from the RTD calibration, wherein a linear regression model correlates heater temperature with resistance. The representative values of the standard error in calculation of the slope and the intercept are 0.04% and 0.05%, respectively. The uncertainties in measurement of the temperature and the heat flux at the heater are given by the following equations, respectively:

$$\left(\frac{\Delta T_{hs}}{T_{hs}} \right)^2 = \left(\frac{\tilde{m}\Delta R_{hs}}{T_{hs}} \right)^2 + \left(\frac{R_{hs}\Delta \tilde{m}}{T_{hs}} \right)^2 + \left(\frac{\Delta \tilde{c}}{T_{hs}} \right)^2 \quad (15)$$

$$\left(\frac{\Delta \dot{q}_{hs}''}{\dot{q}_{hs}''} \right)^2 = \left(\frac{\Delta V_{pr}}{V_{pr}} \right)^2 + \left(\frac{\Delta V_{hs}}{V_{hs}} \right)^2 + \left(\frac{\Delta R_{pr}}{R_{pr}} \right)^2 + \left(\frac{\Delta A_{hs}}{A_{hs}} \right)^2 \quad (16)$$

In summary, the uncertainties in the measurement of heater temperature and heat flux are found to be less than 2% and 1%, respectively.

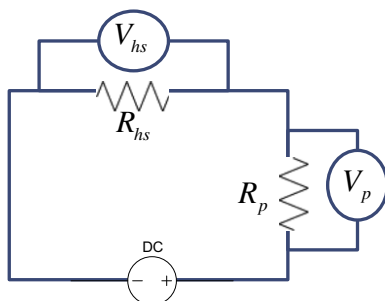


Fig. 4. The electric circuit diagram illustrating the use of precision and heater resistances to calculate the temperature and heat flux.

5. Results and discussion

Applications such as thermal management of electronics demand efficient cooling or dissipation of high heat fluxes at a fixed (low) surface temperature, while water purification systems such as the sweep-gas membrane-distillation (SGMD) require maximum throughput of the distillate. Although it is beneficial to maximize both the heat and mass transfer, it will be shown how certain operating conditions might preclude the occurrence of both effects simultaneously. In other words, since the net heat generated is dissipated via multiple cooling mechanisms, an enhancement in the overall cooling may not correspond to a proportional increase in the throughput of the distillate.

The two critical operating conditions, which can be easily modulated in the application of thin-film evaporation are the flow rates of liquid and air. Factors pertaining to the geometry and material properties of the micro scale device enabling thin film evaporation also affect the overall performance. These include the characteristic liquid film-thickness, the characteristic pore size, porosity, thickness and thermo-physical properties of the membrane. The effect of these factors have been analyzed and presented elsewhere [13,24].

5.1. The effect of liquid flow rate

5.1.1. Computational study of the effect of liquid flow rate

The net rate of evaporation and heat dissipated at different liquid flow rates for a fixed air flow is shown in Fig. 5. In order to determine the performance sensitivity to varying liquid rates, the location of the liquid vapor interface is held constant at the membrane outlet, which eliminates vapor diffusion within the membrane pores. Mathematically, this constraint is enforced by setting $\delta_v = 0$. The effect of liquid flow is studied at a fixed average jet velocity of 30 m/s and a variable inlet velocity of the liquid between 0.001 and 0.004 m/s. This corresponds to Reynolds numbers for liquid flow, Re_l , ranging from 0.25 to 1, where the Reynolds number is calculated as $Re_l = u_l d_h / \nu_l$. The hydraulic diameter for the thin film region is $d_h \sim 10 \mu\text{m}$.

For relatively small flow rates ($Re_l \sim 0.1$), the heat dissipated due to conduction across the film thickness is more efficient than advection or capacitive heating of the liquid flowing between the membrane and the heated source. In this case, heat dissipation is primarily due to evaporation. As the temperature of the heat source approaches saturation condition, the rate of evaporation and cooling are both enhanced. For an inlet velocity of 0.001 m/s, the net heat flux dissipated at 360 K is 104 W/cm². With a quadru-

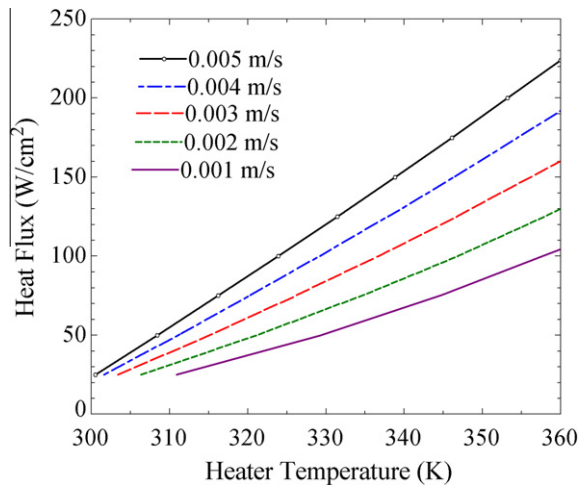


Fig. 5. Net heat flux dissipated versus heater temperature as a function of liquid flow rate ($Re_l \sim 0.25$ – 1.25) for a microscale device illustrated in Fig. 1. The interfacial location is fixed at the membrane outlet, $\delta_v = 0$, and the average velocity of air exiting the nozzle is 30 m/s corresponding to $Re_j \sim 960$.

ple increase in liquid flow rate, the rate of cooling can be enhanced to 192 W/cm², a two-fold increase.

For these operating conditions, the relative contributions of evaporation, convection and spreading to net heat dissipation from the heater are illustrated in Fig. 6a and b. Note that the location of

the interface was fixed at the membrane outlet, which eliminates the resistance to vapor diffusion. At a low liquid flow rate ($Re_l \sim 0.25$), evaporation and sensible cooling contribute about 90% and 7% of net heat dissipation, respectively, and the remaining 3% is attributed to spreading in the substrate. For a quadruple increase in liquid flow rate ($Re_l \sim 1$), the relative contribution of single phase convection is about 34%, which is still lower than the contribution of evaporation of about 64%. The pressure drop to pump liquid across the thin film region for inlet velocities of 0.001 m/s and 0.004 m/s is 6.1 kPa and 24.7 kPa, respectively, which translates to a sixteen-fold increase in pumping power for a quadruple increase in liquid flow rate. At a significantly higher liquid flow rate, $Re_l \sim 2.5$, sensible cooling evidently supersedes contribution from evaporation Fig. 6c. The results demonstrate the enhancement in net heat flux due to a higher contribution from single phase convection at higher liquid flow rates. In applications requiring cooling of small heat sources, the utilization of enhanced advection is viable since the net pumping power is practically achievable. However, for larger heat sources and applications such as membrane distillation, it is necessary to induce significant evaporation and rely less on capacitive cooling, which requires maximizing the rate of vapor transport from the liquid–vapor interface.

In order to compare mass transfer resistances due to vapor advection and diffusion, the net heat flux supplied at the heater is also determined by assuming an efficient advection mechanism, while introducing the maximum resistance to vapor diffusion. Mathematically, this constraint is enforced by considering the membrane pores filled with air vapor mixture only and setting

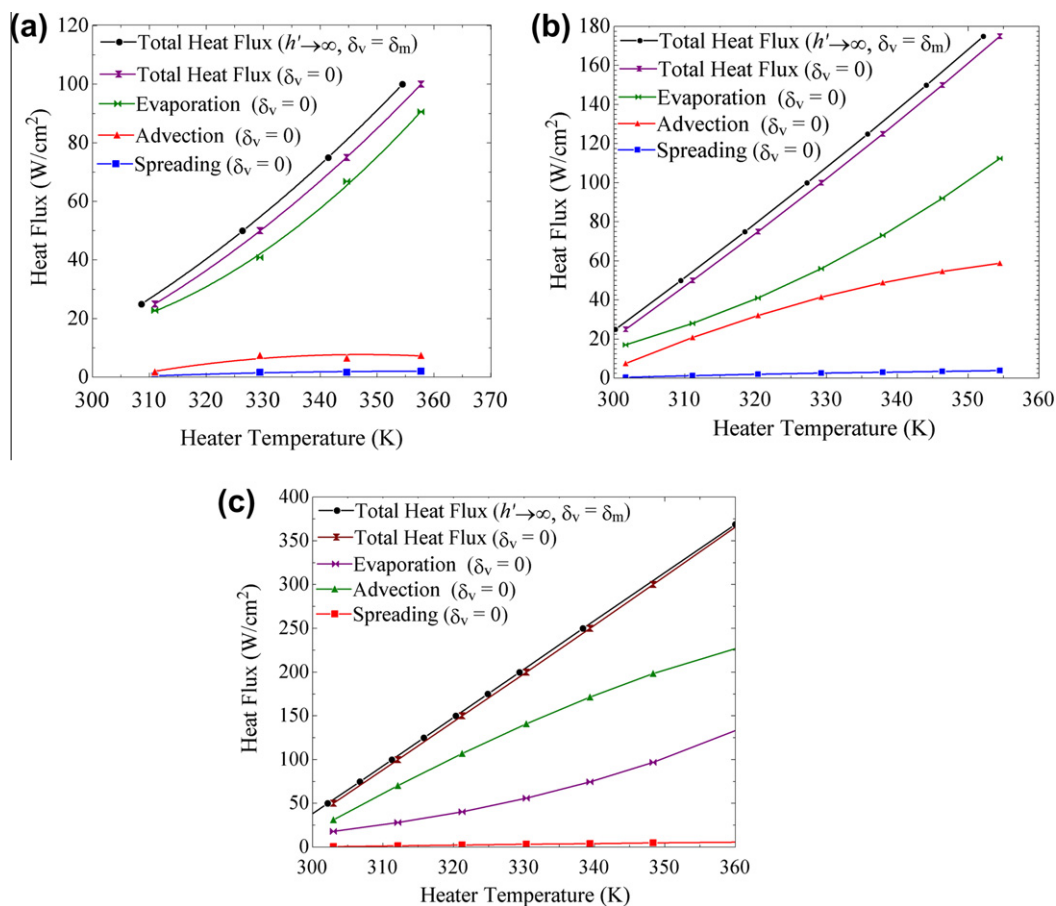


Fig. 6. Relative contributions of evaporation, sensible cooling and substrate spreading to the net heat dissipation from the heated surface. The coolant velocity was maintained at, (a) 0.001 and (b) 0.004 m/s and (c) 0.01 m/s, corresponding to $Re_l \sim 0.25$, 1 and 2.5, respectively. The air velocity was fixed at 30 m/s, corresponding to $Re_j \sim 960$, based on jet diameter. Interface locations at the membrane outlet and inlet are denoted by $\delta_v = 0$ and $\delta_v = m$, respectively; h' denotes the mass transfer coefficient.

$\delta_v = \delta_m$ and $h' \rightarrow \infty$. The net heat flux dissipated at the heated surface for these constraints are also shown in Fig. 6a–c. It is clear that the variation in heat flux is negligible. In other words, by eliminating the advection resistance and introducing vapor diffusion results only in marginal improvement in the net heat flux. This indicates one of the two possibilities; the resistances to vapor diffusion and advection due to the sweeping air are comparable to each other, or the net evaporation is liquid film heat conduction/convection limited. A comparison of different resistances will be shown in the following sections to elucidate this further.

5.1.2. Experimental study of the effect of liquid flow rate

A microscale device with film and membrane thicknesses of 15 μm and 10 μm , respectively, was tested at different liquid flow rates, corresponding to inlet velocities of the liquid between 0.01 m/s and 0.03 m/s. The air velocity was fixed at 65 m/s, for normally impinging dry nitrogen gas from a circular nozzle 1000 μm in diameter, placed 1500 μm from the membrane surface with lateral dimensions of 800 $\mu\text{m} \times 800 \mu\text{m}$.

The net heat flux recorded during the experiments, as a function of temperature of the surface is compared with the computational predictions for the matching operating conditions. The computational results have been evaluated assuming a membrane fully saturated with liquid, $\delta_v = 0$. The inclusion of vapor diffusion resistance in the analysis introduces a negligible variation in the predicted values of device performance due to the dominance of sensible cooling. The results obtained from computational analysis for different liquid flow rates are shown along with experimental observation in Fig. 7. It is apparent that by combining evaporation and sensible cooling modes, heat flux in excess of 500 W/cm² is realizable with heater temperatures under 360 K. In this case, with $Re_l > 2.5$, the contribution of sensible cooling is in excess of 65% as already shown in Fig. 6c.

5.2. The effect of air flow rate

5.2.1. Computational study of the effect of the air flow rate

For a microscale device illustrated in Fig. 1, the net heat flux dissipated at different flow rates for air jet impingement is shown in Fig. 8. The variation in air flow rate at the membrane outlet corresponds to the jet Reynolds number, Re_j between 320 and 1600,

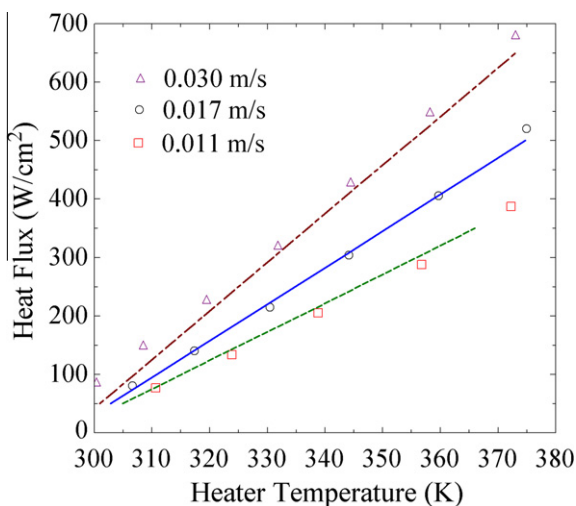


Fig. 7. Net heat flux dissipated at the heater for liquid flow rates much larger than the rate of evaporation ($Re_l > 2.5$). The sweeping air velocity was held constant at 65 m/s, corresponding to $Re_j \sim 4300$ based on the air jet diameter at 20 °C. The computational simulations assume an interfacial location pinned at the membrane outlet ($\delta_v = 0$).

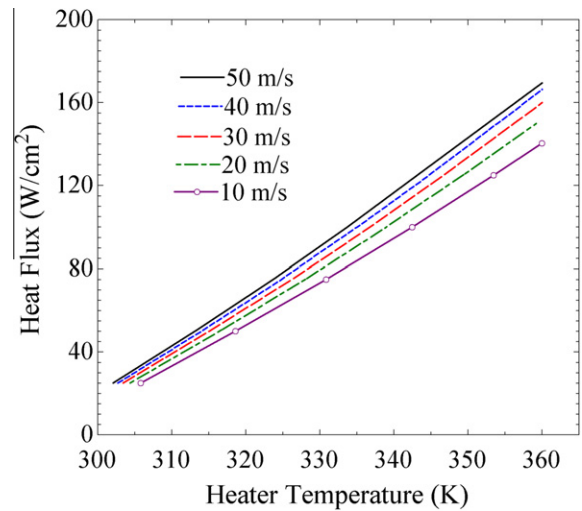


Fig. 8. The net heat flux dissipated versus heater temperature as a function of average air jet velocity at the nozzle exit. The sweeping air velocity was varied between 10 and 50 m/s, corresponding to Re_j between 320 and 1600, based on the air jet diameter at 20 °C. The inlet liquid coolant velocity was held constant at 0.003 m/s corresponding to 20 °C. The computational simulations assume an interface pinned at the membrane outlet ($\delta_v = 0$).

based on the nozzle diameter, evaluated at 20 °C. Contrary to the effect observed by increasing the liquid flow rate, the incremental change in performance is not as significant for a commensurate increase in air flow rate. In other words, a five-fold increase in air velocity from 10 m/s to 50 m/s demonstrates an enhancement of only 20% in the net heat dissipation. In this case, since the contribution from the sensible heating of the liquid remains invariant, the enhancement in rate of evaporation due to the variation in vapor advection can be determined. With a five-fold increase in the flow rate, a 38% increase in the contribution of evaporation to net heat dissipation is observed, although for these conditions the heat and mass transfer coefficients at the stagnation zone are enhanced 2.5 times (see Supplementary information). Clearly, the same extent of improvement in evaporative cooling is not observed, which can be explained based on the combined effect of mass transfer coefficient, vapor pressure and interfacial temperature across the membrane.

In a membrane fully saturated with the evaporating liquid, the interfacial resistance to evaporation per unit area, $(2 - \bar{\sigma}) / (2\bar{\sigma}) \cdot (2\pi RT_{vi}/M_l)^{1/2} / (p_{atm}\epsilon)$ is negligible in comparison to the vapor advection resistance, $RT_{vi} / (M_l p_{atm} \epsilon h')$. For instance, assuming an accommodation coefficient, $\bar{\sigma} = 1$ and porosity, $\epsilon = 0.4$, the interfacial resistance to evaporation for water at $T_{vi} = 300$ K is only 0.012 m² s/kg. On the other hand, the resistance to vapor transport at the stagnation zone corresponding to an air jet velocity of 50 m/s is 1.5 m² s/kg. Therefore, from Eq. (9) it is clear that the rate of evaporation varies as $h'(p_{atm} X_{v,eq}) / T_{vi} = h' p_{v,eq} / T_{vi}$, where the expression within the parenthesis denotes saturation pressure $p_{v,eq}$ at temperature T_{vi} . The variation of $h' p_{v,eq} / T_{vi}$ over the surface of the membrane for different mean velocities of air jet is shown in Fig. 9a.

Although, the mass transfer coefficient, h' increases by 150%, the increase in the average value of $h' p_{v,eq} / T_{vi}$ over the surface of the membrane is only 38%, which matches exactly with the observed increase in the net evaporative heat flux. Although enhanced mass transport can improve vapor advection, a consequent decrease in the interfacial temperature and vapor pressure can depreciate the net rate of evaporation. It is the combination of the three factors that determines the net heat dissipation and distillate throughput for water purification. In order to improve the net rate

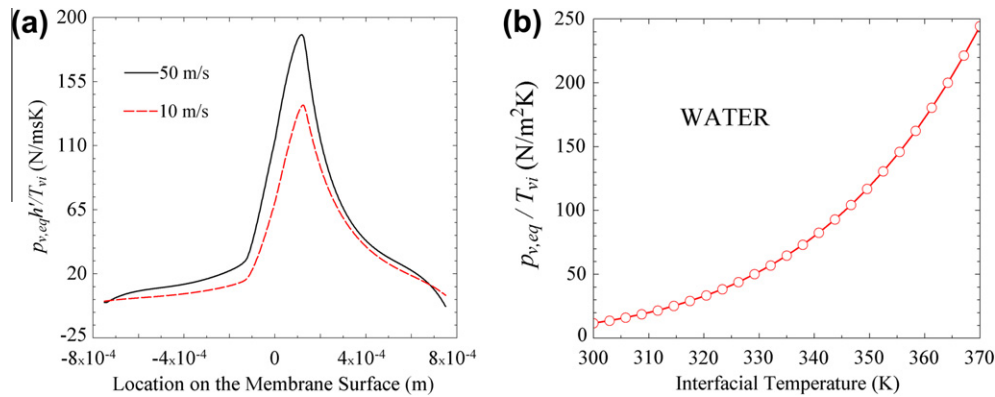


Fig. 9. (a) The variation of $h'p_{v,eq}/T_{vi}$ on the surface of the membrane corresponding to mean air jet velocities of 10 and 50 m/s ($Re_j = 320$ and 1600, respectively). The inlet coolant velocity was held constant at 0.003 m/s corresponding to $Re_l = 0.75$. The computational simulations assume an interfacial location pinned at the membrane outlet ($\delta_v = 0$). (b) Variation of $p_{v,eq}/T_{vi}$ as a function of temperature, T_{vi} for water.

of evaporation, it is essential to utilize a liquid with a large evaporation potential, $p_{v,eq}/T_{vi}$ and a highly efficient vapor advection mechanism providing a large mass transfer coefficient, h' . It can be seen that for water, a significantly higher $p_{v,eq}/T_{vi}$ can be achieved if the heater is operated closer to the saturation temperature, as shown in Fig. 9b. It should be noted that a higher interfacial temperature T_{vi} is ensured by minimizing the thermal resistance to conduction/convection across the liquid film.

5.2.2. Experimental study of the effect of sweep air flow rate

For device features illustrated in Fig. 1, the inlet velocity for the liquid was set at 0.004 m/s, $Re_l = 1$, and the device performance was studied for an average air jet velocity of 30 m/s and 50 m/s from a normally incident air jet of diameter 500 μ m, anchored at a distance of 500 μ m, from the membrane surface. The net heat flux dissipated at the heated surface is plotted as a function of heater temperature as shown in Fig. 10. Corresponding to these operating conditions, the heat flux was also predicted using the computational analysis for interface positions coinciding with the membrane inlet ($\delta_v = \delta_m$) and the outlet, ($\delta_v = 0$). The predicted heat flux corresponding to interface locations at membrane inlet and outlet are shown in Fig. 10a and b. For both interface locations, with the sweep air velocities corresponding to 30 and 50 m/s, the net heat flux dissipated from the heated surface at 360 K was found to be close to 180 W/cm². The net contributions of evaporation and capacitive cooling are predicted as 65% and 35%, respectively, with negligible spreading into the substrate.

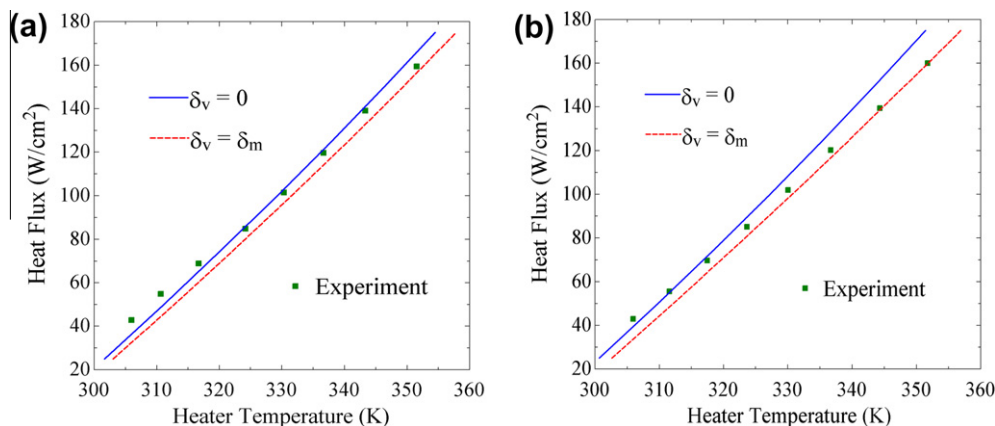


Fig. 10. The net heat flux dissipated versus temperature at the heater. The computational simulations assume interfacial locations pinned at the membrane outlet ($\delta_v = 0$), and inlet ($\delta_v = \delta_m$). The average liquid velocity is 0.004 m/s, corresponding to $Re_l = 1$ and the air jet velocity is (a) 30 m/s, $Re_j = 960$ and (b) 50 m/s, $Re_j = 1600$.

A comparison of experimental results with computational predictions demonstrates an interesting observation. When the liquid flow rate at the inlet is held constant, the net rate of evaporation from the membrane is smaller at lower heat fluxes and it increases as the heater temperature rises with higher heat fluxes. At lower heat fluxes, the experimental observations match well with computational predictions corresponding to the interface pinned at the membrane outlet ($\delta_v = 0$). On the other hand, at higher heat fluxes, they match the predicted heat flux corresponding to the interface located at the membrane inlet ($\delta_v = \delta_m$). This variation is indicative of the system self-adjusting its overall thermal resistance to match the heat flux supplied at the heated surface.

5.3. Resistances limiting net heat and mass transfer

The temperature variation along the heated surface and the membrane inlet is shown in Figs. 11 and 12, respectively. A net heat flux of 100 W/cm² is dissipated at the heater by the combined contribution of evaporation and convection. The resulting temperature distribution at steady state is estimated for different sweep air and liquid film flow rates. The asymmetric distribution is due to the liquid flow, which is supplied at ambient temperature through the inlet port (at the left, Fig. 1a) and gradually heated flowing downstream (right, Fig. 1a) by the heat source. At higher liquid and air flow rates, a lower heater and membrane temperature are expected due to enhanced convection and evaporation, respectively.

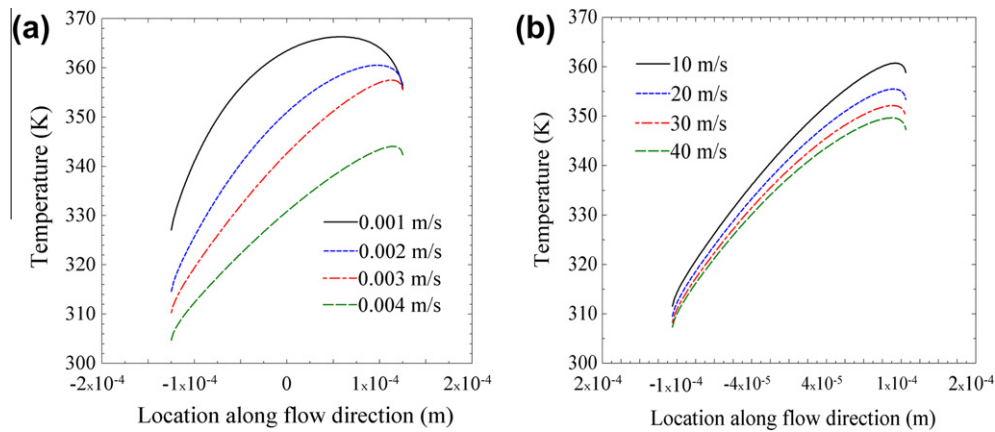


Fig. 11. The temperature along the heated surface is shown for (a) a fixed air jet velocity of 30 m/s, $Re_j = 960$ and variable liquid inlet velocities corresponding to Re_l between 0.25 and 1.25, and (b) a fixed liquid inlet velocity of 0.003 m/s, Re_l , and variable air jet velocities, corresponding to Re_j between 320 and 1280. The computational simulations assume an interfacial location pinned at the membrane outlet ($\delta_v = 0$).

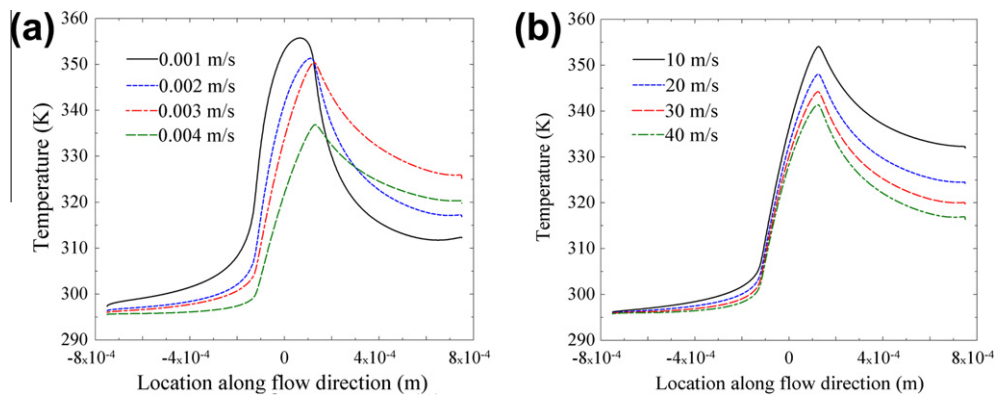


Fig. 12. The temperature along the membrane inlet is shown for (a) a fixed air jet velocity of 30 m/s, $Re_j = 960$ and variable liquid inlet velocities corresponding to Re_l between 0.25 and 1.25, and (b) a fixed liquid inlet velocity of 0.003 m/s, Re_l , and variable air jet velocities, corresponding to Re_j between 320 and 1280. The computational simulations assume an interfacial location pinned at the membrane outlet ($\delta_v = 0$).

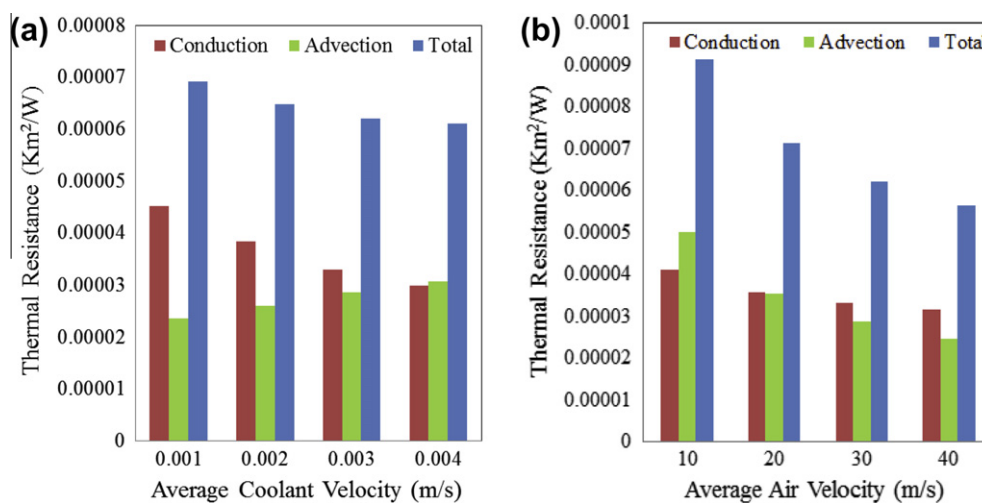


Fig. 13. Thermal resistance corresponding to conduction and vapor advection for (a) a fixed air jet velocity of 30 m/s, $Re_j = 960$ and variable liquid inlet velocities corresponding to between 0.25 and 1.25, and (b) a fixed liquid inlet velocity of 0.003 m/s, Re_l , and variable air jet velocities, corresponding to Re_j between 320 and 1280. The computational simulations assume an interfacial location pinned at the membrane outlet ($\delta_v = 0$).

Based on the net heat dissipated by evaporation alone and the temperature variation across the heated surface (Fig. 11) and membrane inlet (Fig. 12), the thermal resistances to heat conduction

across the liquid film and vapor advection at the membrane outlet are calculated for the device illustrated in Fig. 1. Fig. 13a and b compare the net thermal resistance to conduction and advection

resistance, for different liquid and air flow rates, respectively. The net thermal resistance decreases for a higher liquid and air flow rate. It is to be noted that the decrease in net resistance at a higher liquid and air flow rate is mainly due to a corresponding decrease in thermal resistance due to liquid and air convection, respectively. More importantly, the thermal resistance to convection across the liquid film and vapor advection at the membrane outlet are comparable. Therefore, the overall rate of evaporation can be increased by decreasing the film thickness or utilizing a highly conductive porous structure within the thin-film region along with a more efficient vapor advection mechanism. While the film thickness can be easily controlled by altering the fabrication process [24], it is however more challenging to enhance the overall rate of vapor advection beyond that achievable using jet impingement of dry air. For a cooling mechanism based on evaporation wherein the thickness of the liquid film can be easily minimized, the rate of vapor advection eventually limits the net heat flux and distillate throughput.

6. Conclusion

Efficient cooling and sweep gas membrane distillation of water require maximizing both heat and mass transfer, which can be achieved using evaporation of thin liquid films. Evaporation can be sustained using sources generating heat, which can be electronic components, industrial waste heat or solar power. Evaporation can be maximized by incorporating two key ideas: (1) maintaining a very thin liquid film by capillary confinement using a nanoporous membrane or utilize a conductive porous structure within the thin film region, and (2) using an enhanced vapor advection mechanism like jet impingement of dry air.

An approach demonstrating an innovative and robust technique to control the thickness of the liquid film is presented. Unlike other mechanisms that rely on flow hydrodynamics that typically result in thick films and dryout conditions, capillary confinement can provide precise control of the film thickness on a micrometer and sub-micrometer scale. A microscale device enabling a membrane/capillary-confined, thin-film evaporation is presented along with its performance analysis using a detailed computational study and experimental testing. The analysis explains the overall performance under different operating conditions, and the conclusions are found to be generally applicable to two-phase cooling utilizing thin film evaporation. The overall heat dissipation as a function of two operating conditions, namely the liquid and the air flow rates is evaluated. In order to elucidate device performance, the contributions of evaporation, single phase convection and spreading are discussed at different operating conditions.

The overall heat and mass transfer is found to be greatly affected by varying the liquid flow rate. At flow rates matching the net rate of evaporation from the membrane, the overall heat dissipated at the heated surface is due mostly to evaporation. With increase in the liquid flow rate, the contribution of advection increases dramatically to dissipate higher heat load. The net heat flux at a surface temperature of 360 K is nearly doubled from ~100 to 200 W/cm² for a quadruple increase in flow Reynolds number, Re_l from 0.25 to 1. Even larger increase in heat dissipation is observed for relatively large liquid flow rates. This enhancement in overall cooling capability, due primarily to increased advection is achieved at the expense of higher pumping power for liquid flow. For $Re_l > 2.5$, heat fluxes exceeding 500 W/cm² are obtained at sub-saturation heater temperatures close to 360 K. Although higher heat fluxes are suitable for applications such as electronic cooling, non-optimal evaporation in this mode of operation is not favorable for thermal management of large heat sources or for distillation of water.

The variation in the net rate of evaporation and heat transfer by varying the air flow rate was not found to be significantly large. A five-fold increase in flow rate of air results in only 20% enhancement in the net heat dissipation. The rate of evaporation was found to vary with $h'p_{v,eq}/T_{vi}$, where $p_{v,eq}$ denotes saturation pressure of liquid coolant at temperature T_{vi} . In order to improve the net rate of evaporation, it is essential to utilize a liquid with a large evaporation potential, $p_{v,eq}/T_{vi}$ and a highly efficient vapor advection mechanism provided by a large mass transfer coefficient, h' . For water, a significantly higher $p_{v,eq}/T_{vi}$ can be achieved if the heated surface is operated closer to the saturation temperature by minimizing conduction/convection resistance across the evaporating liquid film.

Based on the net heat dissipated by evaporation alone and the temperature variation across the liquid film and membrane, the thermal resistances to heat conduction across the liquid film and vapor advection at the membrane outlet were calculated. The decrease in net thermal resistance at higher liquid and air flow rates is mainly due to a corresponding decrease in thermal resistance due to conduction and advection, respectively. For the specific cooling device configuration investigated in this work, these resistances are comparable and the overall rate of evaporation can be increased by further decreasing the film thickness to 1 μm or utilizing a highly conductive porous structure within the thin-film region along with a more efficient vapor advection mechanism.

For a specific application to electronics thermal management, the essential requirement for the cooling method is to minimize heat sink volume and to maximize heat dissipation per unit area. Therefore, reducing the thickness of the liquid film further to 1 μm with an efficient vapor advection mechanism can further improve the cooling rate and volumetric density. On the other hand, for membrane distillation, the net rate of evaporation can be greatly enhanced if the liquid film region incorporates a highly conductive porous material over a much larger area of heat source. This modification would allow the use of thick films with lesser pumping power, but without compromising the overall system performance.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ijheatmasstransfer.2012.11.015>.

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